

Ecosystem Indicator Workflows Development Plan

Requirements Gathering and Two-Year Development Roadmap

Ecosystem Indicators Workflows Project Team

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Executive summary

The Ecosystem Indicator Workflows (EIW) project aims to establish a sustainable framework for the development, sharing, and reuse of ecosystem-specific indicators through documented, reproducible, and modular computational workflows. The project responds to a growing need for transparent and scalable approaches to ecosystem assessment, monitoring, reporting, conservation planning, ecosystem accounting, and environmental decision-making.

Ecosystem indicators are increasingly used to understand ecosystem condition, measure ecological change, support environmental reporting, and track progress towards national and international biodiversity objectives. However, the methods used to develop these indicators are often developed independently for specific projects, resulting in duplicated effort, limited reproducibility, inconsistent documentation, and challenges in adapting workflows to new ecosystems and applications.

The EIW project addresses these challenges through the development of a common workflow framework that supports the creation of fit-for-purpose ecosystem indicators. Each workflow is designed to begin with a user-defined scope, including the definition of a target ecosystem, intended purpose, diagnostic mechanism, and indicator variable. Workflows follow a consistent structure comprising ecosystem data preparation, indicator data preparation, indicator calculation, and validation and uncertainty assessment. To promote scientific consistency and interoperability, ecosystem definitions are aligned with the IUCN Global Ecosystem Typology and workflows are designed to follow FAIR and reproducible research principles.

This Development Plan presents the outcomes of the project’s requirements gathering activities and defines a two-year roadmap for workflow development. Requirements were collected through stakeholder workshops, consultations, interviews, and engagement activities involving researchers, ecologists, software developers, data scientists, environmental managers, government agencies, infrastructure providers, and other prospective users of ecosystem indicator workflows. These activities identified strong demand for workflows that are adaptable, transparent, interoperable, well-documented, and easy to reuse across different ecosystem types and assessment contexts.

The plan identifies nine primary user groups and a series of priority use cases spanning indicator development, environmental reporting, ecosystem accounting, environmental management, workflow publication, workflow discovery, training, and nature-related reporting applications. These use cases provide the basis for prioritising workflow functionality and establishing a staged delivery roadmap.

The roadmap is structured around two development tranches. The first tranche focuses on establishing the foundational components of the workflow ecosystem, including a modular workflow architecture, ecosystem and indicator data preparation workflows, data harmonisation approaches, validation methods, demonstration workflows, metadata standards, and introductory training resources. The second tranche focuses on scaling and operationalising the workflow ecosystem through the development of workflow catalogues, enhanced interoperability services, vocabulary crosswalks, FAIR-compliant workflow publication, advanced training resources, and expanded workflow coverage.

The Development Plan also establishes a framework for ongoing review and prioritisation. Functionality planned for the second development tranche will be reviewed during the fourth quarter of Year 1 to ensure that future development continues to reflect user needs, technological opportunities, and emerging ecosystem assessment requirements. This adaptive approach recognises that user requirements, data availability, policy demands, and scientific methods will continue to evolve throughout the life of the project.

Ultimately, the EIW project seeks to build more than a collection of workflows. It aims to establish a community-supported toolbox of reusable methods, software, data resources, and training materials that enable ecosystem indicators to be developed, shared, adapted, and maintained across organisations, disciplines, and jurisdictions. By embedding principles of reproducibility, transparency, interoperability, and reuse throughout the workflow lifecycle, the project will support more consistent and accessible ecosystem indicator development for research, management, policy, and reporting applications.

Document Details

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Revision history

Version	Date	Editor	Summary of Changes
0.1	April 2026	Abhimanyu Raj Singh, Silvia Colombo	Initial draft based on WP2 planning and requirement specifications
0.2	June 2026	Abhimanyu Raj Singh	Updated draft of User Groups, User stories and User requirements after first workshop
0.3	6 July 2026	Abhimanyu Raj Singh, Silvia Colombo, Ada Sánchez-Mercado, [AI tools: Claude]	Updated User Groups, User stories and User requirements, incorporated human and AI generated summaries from both workshop, consolidate documents with multiple sections
0.4	10 July 2026	JR Ferrer-Paris, [AI tools: Co-pilot]	Consolidate information from multiple documents, provide document structure assisted by AI-tools, use of markdown authoring and version control tools
0.5	20 July 2026	JR Ferrer-Paris	Review, improve, edit or remove AI-generated content.
0.6	21 Aug 2026	JR Ferrer-Paris, Abhimanyu Raj Singh, Ada Sanchez-Mercado [AI tools: Co-pilot]	Introduction with overview and scope. Restructuring user groups, requirements and use cases after whiteboard session. Review of AI-generated content.
0.7	30 Sep 2026	JR Ferrer-Paris, Paula Martinez, Elliot Gould, Yannick Tidou [AI tools: Co-pilot]	Summarise and address feedback on use cases, add sustainability section, add executive summary

Part I

Part I - Context

1 Introduction

1.1 Background

Ecosystem-specific indicators are used to support ecosystem assessment, monitoring, reporting, conservation planning, ecosystem accounting, and environmental decision-making. These indicators provide a structured way to measure ecosystem characteristics, assess ecological condition, track change through time, and evaluate progress towards management and policy objectives.

Despite their growing importance, the development of ecosystem indicators is often hampered by fragmented analytical approaches, inconsistent methods, poorly documented workflows, and challenges associated with integrating diverse data sources. Indicator development frequently relies on bespoke analyses that are difficult to reproduce, adapt to new contexts, or apply consistently across ecosystem types. These limitations reduce transparency, constrain scalability, and hinder the uptake of ecosystem indicators by the broader research, policy, and management communities.

The Ecosystem Indicator Workflows (EIW) project was established to address these challenges through the development of modular, transparent, and reproducible workflows that support the creation of ecosystem-specific indicators. The project aims to enable users to develop fit-for-purpose indicators by combining ecological expertise, data resources, computational workflows, and reproducible research practices within a common framework.

1.2 FAIR workflows for Ecosystem-specific Indicators

For the purpose of this document an ecosystem indicator workflow is a documented and reproducible computational workflow for the development of ecosystem-specific and fit-for-purpose indicators. Each workflow begins with a clearly defined scope and follows a common structure that supports transparency, consistency, and reuse.

The workflow framework comprises four core components:

1. Ecosystem data preparation;
2. Indicator data preparation;
3. Indicator calculation; and
4. Validation and uncertainty assessment.

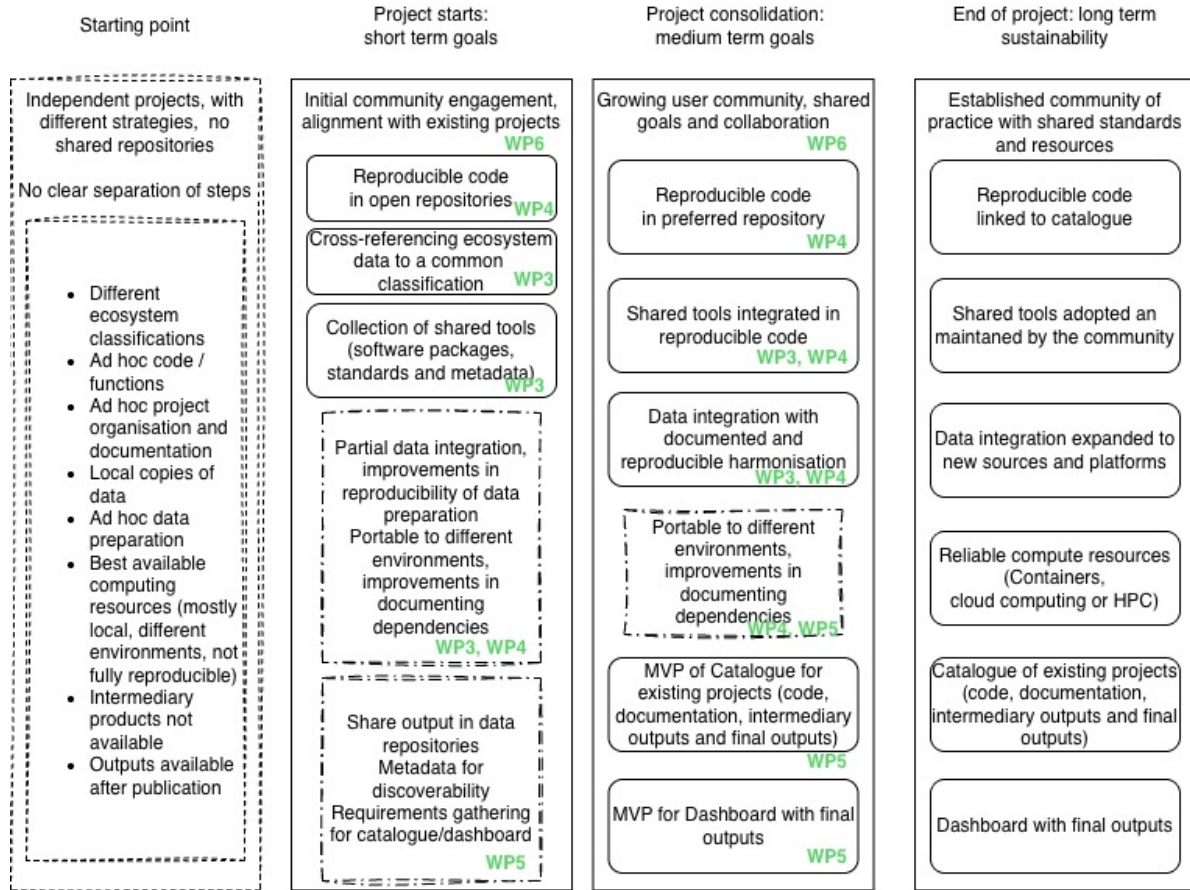


Figure 1.1: Schematic representation of the vision of the EIW project, moving from a current state of fragmented and ad hoc methods towards fully reproducible workflows that are supported by a community of practice and that integrate seamlessly into online catalogues and reporting tools. The coloured codes in each box refer to specific work packages of the EIW project plan. WP3: Work Package 3 - Ecosystem Data and Metadata; WP4: Work Package 4 - Workflow Delivery; Work Package 5 - Indicator Workflow Catalogue and Dashboard; Work Package 6 - Community Building and Engagement, Communication, Skills and Training. Diagram available in draw.io

The scope of each workflow is defined through:

1. Identification of the target ecosystem;
2. Definition of the intended purpose of the indicator;
3. Description of the diagnostic mechanism linking the indicator to ecosystem condition or change; and
4. Selection of the indicator variable or variables.

To support interoperability and scientific consistency, ecosystem definitions are aligned with the IUCN Global Ecosystem Typology and workflows are designed to incorporate the best available ecosystem and indicator datasets appropriate to the intended purpose.

1.3 Requirements Gathering and Planning

A key objective of the project is to ensure that workflow development is driven by the needs of the communities that will create, adapt, maintain, and use ecosystem indicator workflows. During the first phase of the project, requirements gathering activities were undertaken with researchers, ecologists, data scientists, software developers, infrastructure providers, environmental managers, government agencies, and other potential users of the workflows.

These activities sought to:

- identify user groups and their requirements;
- understand high-level workflow requirements and constraints;
- document priority use cases;
- identify opportunities for workflow reuse and adaptation;
- understand barriers to adoption; and
- inform the prioritisation of workflow functionality over the life of the project.

The findings from these activities provide the foundation for the workflow development roadmap described in this document.

1.4 Purpose of this Development Plan

This document presents a documented workflow development plan for the period 2026–2028. This document describes the requirements gathered from stakeholders and translates them into a staged development roadmap for workflow delivery.

Specifically, the plan:

- identifies the primary user groups and their requirements;
- documents high-level workflow requirements and constraints;

- describes priority use cases for ecosystem indicator workflows;
- defines the functionality to be delivered during the first and second workflow development tranches;
- outlines the process for reviewing and refining future priorities; and
- provides a framework for guiding workflow development, community engagement, and future investment.

The plan will be used by project partners, workflow developers, stakeholders, and governance groups to guide implementation of the Ecosystem Indicator Workflows project and to ensure that workflow development remains aligned with user needs, scientific best practice, and the long-term objective of building a sustainable ecosystem indicator workflow community.

This document focuses only on activities directly related to Work Package 4 - Workflow Delivery. There will be other documents outlining requirements and guidance for other work packages.

2 Requirements Gathering Activities

2.1 Objectives

Requirements gathering activities were undertaken to:

- Identify workflow functionality required by prospective users.
- Develop detailed use cases.
- Understand workflow requirements and constraints.
- Inform prioritisation of the first and second workflow development tranches.

2.2 Engagement Activities

Activity	Date	Purpose
Core team meetings	May - July 2026	General structure and background
Workshop 1	19 May 2026	Scope, users and priorities
Workshop 2	29 June 2026	Workflow requirements and architecture
Interviews / meetings	May–July 2026	Targeted stakeholder feedback
Document review	Continuous	Existing workflows and requirements

2.3 Participants in requirement gathering activities

We had 32 participants in the first stakeholder workshop that involved ecosystem scientists, data analysts, government practitioners, research software engineers, data managers, and platform providers. We had 12 participants in the second workshop that targeted researchers with direct experience in ecosystem assessments. We gathered feedback through participation in 18 meetings with different participants from government, data scientists, platform providers, and researchers and students.

Part II

Part II - User Needs

3 User Groups

The Ecosystem Indicator Workflow (EIW) program serves a diverse community of users with different technical capabilities, responsibilities, and objectives. User groups were identified through stakeholder workshops, requirements gathering activities, market analysis, and community engagement planning.

The user groups described below represent the primary audiences for workflow development, testing, adoption, and long-term sustainability.

ID	User Type	Description	Primary Interaction
UG01	Ecologist / Conservation Scientist	Subject-matter expert responsible for defining ecological concepts, ecosystem boundaries, diagnostic mechanisms, and indicator variables. Typically familiar with ecosystem classification systems such as the IUCN Global Ecosystem Typology and ecosystem assessment frameworks. Technical skills vary considerably.	Defines workflow scope; identifies indicator requirements; reviews assumptions; validates indicator outputs; contributes ecological expertise throughout workflow development.
UG02	Student / Early-Career Researcher	Learner and emerging practitioner using workflows as educational and skill-development resources. Often has limited experience with ecosystem indicators, reproducible workflows, or scientific programming.	Participates in training; runs demonstration workflows; contributes testing and documentation feedback; develops workflow skills through worked examples and guided exercises.

ID	User Type	Description	Primary Interaction
UG03	Environmental Manager / Technical Practitioner	Operational user applying ecosystem indicators to support ecosystem management, conservation planning, monitoring programs, natural resource management, restoration activities, or environmental assessments. May work in government, NGOs, Indigenous organisations, or consultancies.	Executes workflows or uses workflow outputs to support management decisions; provides feedback on usability and operational requirements; contributes use cases and validation examples.
UG04	Environmental Policy Analyst / Reporting Officer	User of ecosystem indicators for policy development, environmental reporting, ecosystem accounting, State of the Environment reporting, biodiversity monitoring, and compliance obligations. Typically consumes workflow outputs rather than running workflows directly.	Accesses indicator outputs, reports, metadata, and provenance records; evaluates indicators for policy and reporting purposes; identifies emerging reporting requirements.
UG05	Environmental Consultant / Assurance Practitioner	Professional user applying ecosystem indicators within environmental consulting, auditing, sustainability reporting, ecosystem accounting, and nature-related disclosure activities. Includes organisations supporting TNFD and related reporting frameworks.	Uses workflow outputs and methods to support client assessments; adapts workflows to project-specific requirements; contributes implementation feedback and industry use cases.
UG06	Data Scientist / Technical Analyst	Technical practitioner who applies, adapts, and extends workflows for research, monitoring, and assessment purposes. Typically uses R, Python, GIS tools, and statistical workflows.	Runs workflows using new datasets; adapts workflows for new ecosystems or assessment purposes; performs data preparation and quality assurance; develops new workflow components.
UG07	Research Software Engineer (RSE) / Workflow Developer	Technical specialist responsible for designing, implementing, maintaining, and documenting workflows. Develops reproducible computational methods, modular workflow components, and software infrastructure.	Develops workflow code; manages repositories; implements FAIR practices; maintains software environments; contributes documentation and workflow standards.

ID	User Type	Description	Primary Interaction
UG08	Data Provider / Data Custodian	Organisation or individual responsible for generating, managing, or publishing ecosystem and indicator datasets. Includes government agencies, monitoring programs, research infrastructures, and repository managers.	Provides authoritative datasets; supports metadata and provenance requirements; advises on data quality, limitations, licensing, and update schedules.
UG09	Platform Administrator / Research Infrastructure Provider	Technical personnel responsible for workflow deployment, hosting, security, interoperability, performance, and long-term infrastructure support. Includes providers of cloud, HPC, and research infrastructure platforms.	Deploys workflow environments; manages containerised infrastructure; supports interoperability, authentication, monitoring, and operational sustainability.
UG10	Science Communicator and Trainer	Translates ecosystem indicator outputs and methods into accessible formats for diverse audiences (educators, outreach officers, training coordinators). Strong in communication/pedagogy; may have limited technical background.	Converts outputs into plain-language summaries, visuals, and training materials; bridges technical developers (UG06/UG07) and non-technical users (UG03/UG04); gathers feedback to improve accessibility.

4 User Requirements for Ecosystem Indicator Workflows

4.1 Overview

Workshop participants explored a series of realistic workflow development scenarios and identified user needs, technical challenges, and priorities for future Ecosystem Indicator Workflows (EIWs).

Workshop participants consistently emphasised that the value of Ecosystem Indicator Workflows extends beyond indicator calculation. Users require workflows that are adaptable, reproducible, transparent, well-documented, and capable of integrating diverse ecosystem datasets while remaining accessible to users with varying technical capabilities.

4.2 Considerations for user requirements

4.2.1 Workflow Adaptability

Users require workflows that can be adapted to different ecosystem types, geographic regions, and assessment purposes without requiring complete redevelopment. Stakeholders identified adaptability as one of the most important requirements for the initiative.

Participants repeatedly highlighted that existing workflows are often difficult to repurpose because ecosystem-specific assumptions and methods are embedded throughout the workflow rather than isolated in configurable components.

4.2.2 Reproducibility and Transparency

Users require workflows that can be independently reproduced, verified, and maintained over time. Reproducibility was identified as a core requirement underpinning scientific credibility and long-term adoption.

Users highlighted challenges associated with legacy workflows, undocumented dependencies, restricted data access, and software version incompatibilities. Participants identified containerisation and explicit environment management as important mechanisms for maintaining reproducibility.

4.2.3 Data Discovery, Access and Integration

Users require workflows that simplify the process of locating, understanding, accessing, and combining ecosystem and indicator datasets from multiple sources. Data preparation was consistently identified as the most time-consuming component of indicator development.

Participants highlighted common challenges including fragmented datasets, inconsistent survey methodologies, varying spatial resolutions, uneven geographic coverage, and difficulties obtaining access to restricted datasets.

4.2.4 Classification and Vocabulary Alignment

Users require ecosystem workflows that support alignment between national ecosystem classifications and international standards.

Stakeholders identified ecosystem classification mapping as one of the largest barriers to workflow development and noted that a substantial proportion of effort is currently spent reconciling classification systems across jurisdictions and assessment frameworks.

4.2.5 Documentation, Training and Knowledge Transfer

Users require documentation and training resources that accommodate varying levels of technical expertise.

Participants consistently identified documentation and training as critical enablers of workflow adoption. Requirements varied across user groups, but all groups emphasised the need for high-quality documentation and practical examples.

4.2.6 Accessibility and Usability

Users require workflows that minimise technical barriers to adoption while supporting a diverse user community.

Workshop participants demonstrated strong familiarity with R, Python, RStudio, QGIS, and Jupyter-based environments, while familiarity with specialised platforms and workflow technologies varied considerably. As a result, workflow design should prioritise accessibility and user experience over technical complexity where practical.

4.2.7 Validation, Uncertainty and Scientific Confidence

Users require workflows that support the transparent evaluation of indicator quality and reliability.

Participants noted that uncertainty information is frequently omitted from final products despite being critical for interpretation and decision-making. Validation approaches and integration of expert judgement were identified as areas requiring greater consistency across workflows.

4.3 Priority User Requirements

Across all workshops, the following requirements emerged as the highest priorities for the Ecosystem Indicator Workflow program:

1. Adaptable and modular workflow architecture.
2. Reproducible workflows supported by version-controlled computational environments.
3. Alignment with the IUCN Global Ecosystem Typology and other recognised standards.
4. Transparent data provenance and metadata.
5. Guidance for integrating heterogeneous ecosystem datasets.
6. Comprehensive documentation and tiered training materials.
7. Explicit treatment of validation, uncertainty, and expert judgement.
8. FAIR-compliant workflows, tools, metadata, and outputs.

Requirements 1 to 7 provide the foundation for prioritising workflow functionality and defining the staged delivery roadmap described in subsequent sections of this document. Requirement 8 is embedded in all activities of the project.

4.4 Key requirements

The following list of key requirements represent a comprehensive set of needs identified across stakeholder groups. They map to the priority requirements, except that the points 5 and 6 are merged for practical reasons.

ID	Description
	<i>AD Adaptability</i>
AD01	Modular workflow designs that clearly separate ecosystem-specific components from generic analytical components.
AD02	Clearly documented points within workflows where ecosystem definitions, data sources, indicator variables, and assessment purposes can be substituted.
AD03	The ability to apply a common workflow structure across multiple ecosystem types.
AD04	The ability to support alternative assessment frameworks, including ecosystem risk, ecosystem condition, recovery, and ecosystem accounting applications.
AD05	Guidance for reconciling global datasets with local and jurisdictional data sources.
	<i>RE Reproducibility and transparency</i>
RE01	Fully documented workflow processes from data acquisition through to final outputs.
RE02	Clear provenance records for all input datasets.
RE03	Version-controlled workflow code and analytical methods.
RE04	Documented software dependencies and computational environments.
RE05	Long-term maintainability as software platforms and packages evolve.
RE06	Repeatable execution across multiple computing environments.
	<i>VO Classification and vocabularies</i>
VO01	Alignment with the IUCN Global Ecosystem Typology.
VO02	Clear mapping between local ecosystem classifications and global typologies.
VO03	Use of authoritative identifiers where possible.
VO04	Machine-readable classification mappings and metadata.
VO05	Transparent documentation of classification decisions and assumptions.
	<i>DD Data discovery</i>
DD01	Guidance on identifying authoritative ecosystem and indicator datasets.
DD02	Clear documentation of data provenance, quality, currency, and fitness-for-purpose.
DD03	Consistent approaches for harmonising heterogeneous datasets.
DD04	Mechanisms to support integration of biological surveys, monitoring data, remote sensing products, and spatial datasets.

ID	Description
DD05	Support for workflows operating across local, cloud, and high-performance computing environments.
DD06	Explicit documentation of data limitations and known biases.
	<i>DO Documentation and training</i>
DO01	Plain-language workflow documentation.
DO02	Worked examples using real ecosystem assessments.
DO03	Step-by-step implementation guidance.
DO04	Technical documentation describing workflow functions and analytical methods.
DO05	Guidance on interpretation of outputs, assumptions, limitations, and uncertainty.
DO06	Training resources suitable for beginner, intermediate, and advanced users.
	<i>AC Accessibility</i>
AC01	Support for commonly used open-source tools and platforms.
AC02	Use of technologies already familiar to the target user community where possible.
AC03	Clear installation and setup instructions.
AC04	Support for users operating within institutional security and software restrictions.
AC05	Outputs that can be interpreted by both technical and non-technical users.
	<i>VA Validation</i>
VA01	Explicit validation procedures.
VA02	Documentation of assumptions and methodological limitations.
VA03	Quantification and communication of uncertainty.
VA04	Support for expert judgement where empirical validation data are unavailable.
VA05	Consistent reporting of confidence in indicator outputs.

5 High-Level Workflow Requirements and Constraints

5.1 Purpose

Ecosystem Indicator Workflows (EIWs) are documented, transparent, and reproducible computational workflows developed to generate ecosystem-specific and fit-for-purpose indicators. The workflows are intended to support consistent indicator development across ecosystem types while providing sufficient flexibility to address different management, monitoring, assessment and reporting needs.

All workflows are designed to align with FAIR principles (Findable, Accessible, Interoperable and Reusable) and support reproducible research practices.

5.2 Scope and Structure

The development of an Ecosystem Indicator Workflow must begin with a clearly defined scope. The scope establishes the context, purpose and intended application of the workflow and provides the basis for all subsequent workflow development activities.

Each workflow must define the following four attributes:

1. ***Target Ecosystem Definition:*** The workflow must identify the ecosystem or ecosystem group that will be assessed.
2. ***Intended Purpose:*** The workflow must clearly state the purpose of the indicator and the decision-making context in which it will be used. Examples may include: Environmental reporting, Natural capital accounting, Ecosystem risk assessment, Ecosystem recovery and integrity assessment, Conservation planning.
3. ***Diagnostic Mechanism:*** The workflow must describe the ecological mechanism through which the selected indicator provides information about ecosystem status, condition or change.
4. ***Indicator Variable Selection:*** The workflow must identify one or more measurable variables that can be used to quantify the selected ecosystem characteristic.

All Ecosystem Indicator Workflows will follow a common modular structure consisting of four core workflow components.

1. ***Ecosystem Data Preparation:*** This module prepares ecosystem spatial data and associated ecosystem information required for indicator calculation. Dataset selection decisions must be documented and justified.
2. ***Indicator Data Preparation:*** This module prepares the data required to quantify the selected indicator variable. All processing steps must be transparent and reproducible.
3. ***Indicator Calculation:*** This module implements the analytical methods required to generate the ecosystem indicator. Calculation methods should be designed to allow consistent application across assessment periods where appropriate.
4. ***Validation and Uncertainty Assessment:*** This module assesses the performance, reliability and limitations of the indicator. Validation outcomes must be documented alongside indicator results.

5.3 Design Principles

The Ecosystem Indicator Workflow program will be guided by the following principles:

1. **Scientific credibility** – indicators are based on sound ecological understanding and defensible methods.
2. **Fitness for purpose** – indicators are developed to address clearly defined user needs and decision contexts.
3. **Transparency** – methods, assumptions and data sources are fully documented.
4. **Reproducibility** – workflows can be repeated and independently verified.
5. **Interoperability** – workflows align with recognised standards and classification frameworks.
6. **Scalability** – workflows can be adapted to additional ecosystems and indicator types over time.
7. **Accessibility** – tools and outputs are usable by a broad community of researchers, practitioners and decision makers.
8. **Continuous improvement** – workflows can be updated as new data, methods and user requirements emerge.

5.4 High-Level Workflow Requirements

5.4.1 Scope Definition Requirements

5.4.2 Workflow Structure Requirements

5.4.3 Software and Technical Requirements

The Ecosystem Indicator Workflow framework must be supported by software tools and computational environments that promote transparency, accessibility and reproducibility. Workflow implementation should minimise barriers to adoption and facilitate long-term maintenance.

This will be achieved through:

- Use of open-access or widely accessible software where feasible.
- Use of documented and version-controlled analytical processes.
- Support for workflow portability across organisations and jurisdictions.
- Ability to adapt workflows for alternative ecosystems and use cases.
- Transparent documentation of software dependencies, packages and computational environments.

5.4.4 FAIR and Reproducibility Requirements

All workflows, tools and outputs must align with FAIR principles and reproducible research practices.

5.4.4.1 Findable

Workflow documentation, code, metadata and outputs should be discoverable through appropriate repositories and documentation systems.

5.4.4.2 Accessible

Workflow components and outputs should be accessible to authorised users through documented and sustainable access mechanisms.

5.4.4.3 Interoperable

Data, metadata and outputs should use recognised standards, formats and classification systems to support integration with other workflows, platforms and reporting systems.

5.4.4.4 Reusable

Workflow documentation must provide sufficient detail to enable independent application, review, adaptation and reuse by other practitioners.

5.5 Constraints

The development and implementation of Ecosystem Indicator Workflows will be subject to several constraints.

- ***Data Availability***

- Indicator development is dependent on the availability, quality and coverage of ecosystem and indicator datasets.
- Data availability will vary across ecosystem types, jurisdictions and time periods.
- Some workflows may require the use of proxy variables where direct measurements are unavailable.

- ***Data Consistency***

- Differences in spatial scale, classification systems, temporal coverage and collection methodologies may affect comparability across datasets.
- Additional data harmonisation may be required to support national-scale assessments.

- ***Computational Constraints***

- Computational methods should balance scientific rigour with practical implementation requirements.
- Workflows should remain maintainable by future users and project teams.

- ***Resource Constraints***

- Workflow development priorities must reflect available funding, staff capacity and stakeholder demand.
- Not all desired functionality may be delivered within the initial development period.

- ***Scientific Constraints***

- Ecological processes are inherently complex and may not be fully represented by individual indicators.
- All indicators should clearly communicate assumptions, limitations and uncertainty associated with interpretation.

Part III

Part III - Use Cases

6 Detailed Use cases

6.1 Overview

Requirements gathering activities identified a range of user groups that will contribute to, operate, or consume Ecosystem Indicator Workflows (EIWs). The use cases below represent the most common and highest-priority applications identified through workshops, user stories, and stakeholder consultations. These use cases informed the prioritisation of workflow functionality and the staged development roadmap.

6.2 Use case descriptions

UC01 – Adapt an existing fit-for-purpose ecosystem-specific indicator to make it reproducible

Attribute	Description
Primary User	Ecologist / Conservation Scientist (UG01)
Supporting Users	Research Software Engineer (UG07), Data Scientist (UG06)
Purpose	Adapt an ecosystem-specific indicator to make it reproducible
Trigger	Update existing ecosystem assessments, revisit research or monitoring questions.
Key requirements	AD01, AD02, AD05, RE01, RE02, RE04, VO01, VO02, DD02, DD03, DD04, DD06, DO04, AC01, AC03, VA02, VA03, VA04, VA05

6.2.0.1 Implementation steps

1. Check existing scope of the workflow as per high-level requirements: ecosystem, purpose, diagnostic and variable selection.

2. Check structure of the workflow as per high-level requirements: data preparation, calculation and validation/uncertainty assessment.
3. Assess existing gaps in implementation of FAIR principles
4. Edit/modify steps to close gaps in FAIR principles
5. Review outputs and document all steps as per high-level requirements.

6.2.0.2 Outputs

- Indicator values
- Maps and visualisations
- Workflow documentation including validation and uncertainty reports

6.2.0.3 Impact

Enable reproducibility of existing ecosystem-specific indicators, increase consistency, transparency, and reproducibility across ecosystem assessments. This use case emerged as a high-priority scenario during workshop discussions around workflow adaptability.

UC02 – Develop a new fit-for-purpose ecosystem-specific indicator

Attribute	Description
Primary User	Ecologist / Conservation Scientist (UG01, UG02)
Supporting Users	Research Software Engineer (UG07), Data Scientist (UG06)
Purpose	Develop an ecosystem indicator for a previously unassessed ecosystem type.
Trigger	A new ecosystem assessment, conservation program, or research project requires an indicator.

6.2.0.1 Implementation steps

1. Define the scope of the workflow as per high-level requirements: ecosystem, purpose, diagnostic and variable selection.
2. Develop structure of the workflow as per high-level requirements: data preparation, calculation and validation/uncertainty assessment.
3. Assess implementation of FAIR principles
4. Review outputs and document all steps as per high-level requirements.

6.2.0.2 Outputs

- Indicator values
- Maps and visualisations
- Workflow documentation including validation and uncertainty reports

6.2.0.3 Impact

Reduces the effort required to develop new indicators while increasing consistency, transparency, and reproducibility across ecosystem assessments. This use case emerged as a high-priority scenario during workshop discussions around workflow adaptability.

UC03 – Adapt an existing reproducible fit-for-purpose indicator to a new ecosystem

Attribute	Description
Primary User	Ecologist / Conservation Scientist (UG01, UG02)
Supporting Users	RSE (UG07), Data Scientist (UG06)
Purpose	Reuse an existing workflow for another ecosystem type or assessment purpose.

6.2.0.1 Implementation steps

1. Locate a suitable workflow in the workflow catalogue.
2. Identify ecosystem-specific components.
3. Replace ecosystem definitions, datasets and parameters.
4. Re-run workflow using the new ecosystem context.
5. Validate outputs.

6.2.0.2 Outputs

- Adapted workflow
- Updated metadata and documentation
- New indicator outputs

6.2.0.3 Impact

Reduces duplication of effort and accelerates ecosystem assessments. Workshop participants consistently identified workflow adaptability as one of the most important requirements for the EIW project.

UC04 – Adapt an existing reproducible ecosystem-specific indicator to a different purpose

Attribute	Description
Primary User	Ecologist / Conservation Scientist (UG01, UG02)
Supporting Users	RSE (UG07), Data Scientist (UG06)
Purpose	Reuse an existing workflow for a different assessment or purpose.

6.2.0.1 Implementation steps

1. Locate a suitable workflow in the workflow catalogue.
2. Identify components related to purpose of the indicator
3. Replace parameters and adjust calculations for the intended purpose.
4. Re-run workflow using the new analytical context and interpretation.
5. Validate outputs.

6.2.0.2 Outputs

- Adapted workflow
- Updated metadata and documentation
- New indicator outputs

6.2.0.3 Impact

Reduces duplication of effort and accelerates ecosystem assessments. Workshop participants consistently identified workflow adaptability as one of the most important requirements for the EIW project.

UC05 – Publish and Maintain a Reproducible Workflow

Attribute	Description
Primary User	Workflow authors (UG01, UG02, UG06, UG07)
Supporting Users	Platform Administrator (UG09)
Purpose	Publish a workflow that can be reused and maintained over time.

6.2.0.1 Implementation steps

1. Package workflow components.
2. Define software dependencies.
3. Configure reproducible environments.
4. Register metadata and identifiers.
5. Publish workflow documentation.

6.2.0.2 Outputs

- Workflow repository
- DOI and metadata records
- Documentation
- Containerised environment

6.2.0.3 Impact

Supports long-term maintenance and reuse while ensuring workflows remain reproducible as software and data sources evolve. Reproducibility and software versioning were identified as critical concerns during workshops.

UC06 – Integrate multiple indicator workflows in assessments or reports

Attribute	Description
Primary User	Ecologist / Technical Analyst (UG01, UG02, UG03)
Supporting Users	RSE (UG07), Data Scientist (UG06)

Attribute	Description
Purpose	Reuse existing workflows and integrate them in a larger ecosystem assessment or report with a defined purpose, including one or multiple ecosystems.

6.2.0.1 Implementation steps

1. Locate suitable workflows in the workflow catalogue.
2. Verify alignment with assessment or report purpose, adapt if necessary (see UC03, UC04)
3. Combine outputs and document selection process
4. Update report metadata including provenance information and references to input workflow (see UC05).

6.2.0.2 Outputs

- Assessment or report with embedded workflows (documented methods and/or outputs)
- Workflow citations

6.2.0.3 Impact

Reduces duplication of effort and accelerates ecosystem assessments. Workshop participants consistently identified workflow adaptability as one of the most important requirements for the EIW project.

UC07 – Discover and Reuse Existing Workflows

Attribute	Description
Primary User	Researchers (UG01, UG02), Technical practitioner (UG03), Policy Analyst (UG04) Consultant (UG05)
Purpose	Find workflows relevant to a specific ecosystem, indicator, or assessment purpose.

6.2.0.1 Implementation steps

1. Search workflow catalogue.
2. Filter by ecosystem type, indicator type, or assessment framework.
3. Review documentation and examples.
4. Execute or adapt workflow (see UC03 and UC04).

6.2.0.2 Outputs

- Reused workflows
- Adapted workflows
- Reproducible assessments

6.2.0.3 Impact

Reduces duplication of effort and encourages workflow standardisation across organisations and jurisdictions.

UC08 – Training and Skills Development

Attribute	Description
Primary User	Student / Early-Career Researcher (UG02)
Supporting Users	Ecologists (UG01), Trainers (UG10)
Purpose	Build capability in reproducible ecosystem assessment workflows.

6.2.0.1 Implementation steps

1. Attend workshops or training events.
2. Access worked examples.
3. Run demonstration workflows.
4. Modify workflows for new examples.
5. Contribute improvements.

6.2.0.2 Outputs

- Trained users
- New contributors
- Community of practice

6.2.0.3 Impact

Expands the user base and supports long-term sustainability of the workflow ecosystem. Training and documentation were identified as high-priority requirements across all workshop participants.

UC09 – Apply an Existing Workflow to a Specific Assessment

Attribute	Description
Primary User	Environmental Consultant / Assurance Practitioner (UG05)
Supporting Users	Environmental Manager / Technical Practitioner (UG03), Data Scientist (UG06)
Purpose	Run an existing, unmodified indicator workflow against a specific site, region, or portfolio to produce a result for an assessment, audit, ecosystem account, or nature-related disclosure.

6.2.0.1 Implementation steps

1. Select a workflow matching the ecosystem and assessment framework required.
2. Confirm the workflow can run within the available data access and IT constraints.
3. Assemble inputs and record data gaps, limitations, and known biases.
4. Execute the workflow; where empirical validation data are unavailable, apply and document expert judgement.
5. Put outputs through documented review before release.

6.2.0.2 Outputs

- Indicator result for the assessment
- Documented limitations statement
- Review and expert-judgement record

6.2.0.3 Impact

Covers routine application of existing workflows, which is expected to be the highest-volume activity across the user community. Distinguishes applied assessment from workflow development (UC01) and adaptation (UC02, UC03), and places a larger focus on documented limitations and expert review as standard requirements. This use case also brings the consulting and assurance community (UG05) into scope, reflecting the growth of ecosystem accounting and nature-related disclosure obligations.

6.3 Summary

Across the requirements gathering process, nine recurring use cases were identified. These can be grouped into four broad categories:

1. **Workflow development and adaptation** (UC01, UC02, UC03, UC04)
2. **Workflow publication, discovery and integration** (UC05, UC06, UC07)
3. **Capacity building and adoption** (UC01, UC08)
4. **Assessment and reporting based on existing indicators** (UC06, UC07, UC09)

These use cases form the basis for prioritising functionality and defining the first and second tranches of workflow development activities described in subsequent sections.

7 Prioritisation Approach

Functionality was prioritised based on four criteria:

Criterion	Description
Strategic Importance	Alignment with project outcomes
Technical Dependency	Required before later functionality
Feasibility	Delivery within available resources
Reusability	Applicability across ecosystems

Priority decisions were informed by workshop feedback, use cases, user stories, project deliverables and partner discussions.

Part IV

Part IV - Roadmap

8 Roadmap Overview

Based on the user groups, requirements and use cases described in previous sections we propose a roadmap for **Work Package 4 Workflow delivery**:

Theme	Use cases	Tranche 1	Tranche 2	Dependencies
Workflow development	UC01 UC02	Priority	Enhance	WP3
Workflow adaptation	UC03 UC04	Explore	Priority	WP3
Workflow publication	UC05	Explore	Priority	
Integration in assessments	UC06	Explore	Explore	
Discoverability	UC07	Initial	Expanded	WP5
Training	UC08	Initial	Expanded	WP6
Applications	UC09		Explore	
FAIR & PID	All	Initial	Full implementation	WP2

Where the first tranche is programed from (May to Nov 2026) and the second tranche (Jan to Nov 2027). This mapping aligns directly with the project's intended outcomes around modular workflows, FAIR workflows, interoperability, workflow catalogues and community adoption.

The proposed roadmap has following dependencies with other project work packages:

- **WP2 PID and FAIR4RS implementation plans** completed during Tranche 1 is required for theme *FAIR & PID*.
- **WP3 Data and Metadata** is completed or in development during Tranche 1 and completed before Tranche 2.
- **WP5 Catalogue and Dashboard** will be developed after completion of Tranche 1 and in parallel to Tranche 2.
- **WP6 Community and Engagement** Training, community building, workshops and user stories are developed in accordance with community and engagement plan.

9 First Tranche (May to Nov 2026)

The first tranche focuses on establishing a minimum viable ecosystem workflow framework that demonstrates adaptability, reproducibility, FAIR compliance and practical applicability across multiple ecosystem indicator use cases. It delivers the foundation required for future workflow expansion while supporting the project's first reporting milestone and initial workflow deployment activities.

The first objective is to use existing indicator examples as a foundation for developing FAIR reproducible indicator workflows (UC01). Second objective is to use the proposed framework to develop new workflows (UC02).

The modular workflow architecture will have documented swap points for ecosystem type, assessment purpose, indicator variables, and data sources allowing initial exploration of adaptation options (UC03 and UC04).

Special attention will be given to initial implementation of FAIR principles and alignment with PID and FAIR4RS implementation plans whenever practicable, aiming at publication of reproducible workflows (UC05).

Integration of workflows into ongoing or planned assessments will be explored (UC06). It is expected that ongoing assessments will be the main drivers and motivation for selecting specific ecosystems and purposes for UC02, UC03 and UC04.

This stage will require an initial exploration of options for discoverability (UC07), coupled with requirement gathering steps for Work Package 5.

The development of this tranche will allow initial opportunities for training (UC08) and involvement of early career researchers and students, with a focus on documentation (introductory workflow documentation), development of training material (beginner-friendly quick-start guide), example workflow notebooks and potentially one community event with published workshop outputs.

10 Second Tranche (Jan to Nov 2027)

The second tranche expands the workflow ecosystem beyond individual workflows to support discoverability, scaling, interoperability and broader community adoption. The focus shifts from workflow creation to workflow reuse and operationalisation.

The main objective is to support and expand adaptation of workflows to new ecosystems and different purposes (UC03 and UC04). Development of new indicator workflows (UC02) will be enhanced and driven by ongoing research projects and assessments.

Workflow development in this stage should follow the FAIR4RS and PID implementation plans, including full publication of workflows (UC05).

Integration of workflows into ongoing or planned assessments will continue to be explored (UC06). It is expected that ongoing assessments will be the main drivers and motivation for selecting specific ecosystems and purposes for UC02, UC03 and UC04.

Scaling up assessment will be coupled with a larger focus on discoverability (UC07) through development of Work Package 5.

The development of this tranche will allow multiple opportunities for training (UC08) and involvement of early career researchers and students. Particularly, the previous experience in development of reproducible indicator workflows from existing workflows (UC01) can be used for didactic purposes. Further development of tiered training materials for beginners, intermediate and advanced users will be complemented with more technical documentation for implementation and maintenance of platform functionality. At least one community event will be completed leading to publication of workshop outputs.

11 Review of Second Tranche Priorities

Consistent with the project plan, second-tranche requirements will be reviewed after the end of Tranche 1 (Nov 2026) and before commencement of Tranche 2 (Jan 2027). The review will evaluate feedback from workflow users, workshop participants, ecosystem experts, and project partners to validate that Year 2 priorities continue to address the most important user needs.

The review will consider:

- User adoption and engagement in the initial phases,
- Technical performance of workflow reproducibility, data interoperability, expected platform performance, and FAIR compliance.
- Community feedback collected through workshops or interviews.
- Steering Committee recommendations

Any approved changes to second-tranche priorities will be documented through the project's governance and change-control processes and reflected in an updated delivery plan.

Part V

Part V - Monitoring and Success

12 Success Indicators

12.1 Year 1 (Nov 2026)

- First tranche workflows delivered.
- At least one workflow demonstrated on EcoCommons.
- Initial training materials available.
- FAIR implementation plans completed.
- User testing completed.

12.2 Year 2 (Nov 2027)

- Second tranche workflows delivered with expanded ecosystem coverage.
- Workflow catalogue (MVP) operational.
- Dashboard (MVP) operational.
- PID implementation completed.
- FAIR workflow publication completed.

12.3 Beyond Project

- Additional workflows contributed by community members.
- Evidence of reuse in ecosystem reporting and management.
- Adoption by government, research and environmental management organisations.

13 Sustainability

EIW's workflows, documentation and training materials need to stay usable and accurate after project funding ends. This section sets out who looks after them, how often they are reviewed, and how responsibility is handed over when the project closes.

13.1 Custodians and Maintainers

Each workflow, documentation set and training resource will have:

- an initial **custodian**: the organisation accountable for the asset;
- at least one **maintainer**: the person or team who keeps it up to date.

Custodians are named as organisations rather than individuals, so accountability isn't lost when staff change. Where possible, each asset has two maintainers, including volunteer maintainers from the user community.

Each asset is labelled **Active**, **Maintenance only** (fixes but no new features) or **Archived** (preserved but no longer updated).

13.2 Maintenance Cadence

- **Monthly**: check that links and identifiers still work, and respond to reported issues.
- **Quarterly**: apply software updates, and test that the example workflows still run.
- **At each release**: update the documentation.
- **Annually**:
 - review the documentation and training materials;
 - confirm each asset's status (Active, Maintenance only or Archived);
- review this plan.
- An extra review is triggered by major software changes, platform changes, or 12 months without any maintenance activity on an asset.

13.3 Governance Handover at Project Close

A **Transition Plan** will be timely prepared before funding ends (e.g. last quarter of funding period), for approval by the governance group.

The Transition Plan will:

- confirm a successor custodian for each asset that stays in use, by written agreement;
- transfer platform accounts (GitHub, Zenodo, EcoCommons and similar) to institutional ownership rather than personal accounts;
- archive assets without a successor, with permanent identifiers so they can still be found and cited;
- deposit a final record of all project identifiers and hand it over to successor organisation, if relevant.

After the project closes, the successor organisation or community steering group carries out an annual review.

Part VI

Appendices

14 Workshop Feedbacks:

14.1 Workshop 1

This workshop was designed as a co-design exercise to gather structured feedback from a heterogeneous, multidisciplinary community of end users. Including ecologists, data analysts, coders, and software engineers, on a demo workflow for ecosystem indicator development, as the first in a series of co-design workshops for the Ecosystem Indicators Workflows (EIW) project aimed at ensuring the workflow reflects how people actually work. The workflow follows FAIR principles and supports applications such as ecosystem risk assessment, ecosystem accounting, and the IUCN Green Status, and given the diversity of expertise, institutions, and working practices among users, the workshop set out to understand how people interact with the workflow in practice, tools, preferred data collections, where friction occurs, and what support and collaboration mechanisms are needed to improve usability and adoption.

Familiarity -> ALA CSIRO FED | shape files tabular and RDA | rstudio GIS, GEE, quarto, Jupyter | R python Bash1

14.2 Workshop 2

This workshop was a second co-design exercise as a follow-up for the first one, bringing together ecologists, data analysts, coders, and software engineers to improve reproducible workflows for ecosystem indicator development. Where Workshop 1 tested and refined a workflow demo against real-world experience, Workshop 2 shifts focus to scoping and reflection: identifying priority ecosystems, biomes, and functional groups to fill urgent knowledge gaps, surfacing which ecosystems/indicators are most useful for current programs versus future scalability, and assessing how reproducible and reusable participants' past indicator-development workflows have actually been.

Ecosystems / focus areas nominated for expanding coverage

- Oyster Reef of Southern & Eastern Australia12%
- Cumberland Plain Woodland, NSW12%
- Mangroves of Moreton Bay8%
- Saltmarshes of Moreton Bay8%
- Alpine snow patch herb fields, SE Australia8%

- Coastal lowland rainforests 8%
- Connected wetlands of the Lake Eyre Basin 8%
- Georgina gidgee woodlands, Central Australia

Planned future work named by participants (next ~2 years)

- Tropical and subtropical rainforests (pending funding)
- Yellow Box–White Box–Blakely’s Red Gum and other woodlands (pending funding)
- Semi-arid and freshwater ecosystems
- Pyric savannas in northern Australia (extending existing Brazil-based indicator work)
- Broader RLE/GSE program: compiling existing assessments and building tools to support new ones at scale

This workshop also asked participants to reflect on a past ecosystem-indicator workflow they had personally worked on. The key findings were:

- Findability: most participants rated finding the needed data as “moderate” (required searching across sources) to “difficult” (scattered/poorly documented); none rated it “very easy.”
- Accessibility: a majority reported data was only “accessible with request/permission” or fully restricted; approx. 71% selected difficulty finding/locating sources as a factor (figure extracted from poll widget — treat as approximate).
- Main accessibility blockers reported: complex access procedures, data ownership issues, broken/outdated links, and low-quality metadata. Free-text comments repeatedly cited data-sharing agreements “taking months” and experts “not responding to emails.”
- Data readiness challenges: inconsistent classification systems (ecosystem types, species names), varying data quality/resolution, temporal mismatches, and difficulty integrating expert knowledge with quantitative data were the most frequently cited.
- Skills required for data preparation: GIS/spatial analysis, remote sensing, and timeseries analysis were most cited, alongside general data cleaning/formatting.
- Reproducibility of the calculation step itself: participants were split between “reproducible” and “partially/difficult to reproduce” — several noted that while code for the quantitative portion was reproducible, the expert-elicitation component (setting thresholds) generally was not documented in a reproducible way.

- Top priority action nominated to improve reproducibility of the existing workflow: “Strengthen documentation” (most-selected option, ahead of standardising methods, validating with independent data, and increasing technical capacity).
- What participants said is needed for the next project to be reproducible from the start: build reproducibility into project design/coding standards from day one rather than as an afterthought; train lead assessors in reproducible coding practices; ensure both accessible data and reproducible compute environments; document species/terminology decisions explicitly.

Topic	Key points	Constraints / challenges	Implications / actions
Workshop 2 outputs	Focus could move beyond single ecosystems to groups of ecosystems to understand how tailored both sets of indicators and individual indicators need to be.	Need to balance general methods across ecosystems with ecosystem-specific adaptation.	Clarify the scale of indicator development: individual ecosystems, ecosystem groups, or cross-ecosystem methods.
Ecosystem focus areas	Tropical and subtropical rainforest, semi-arid woodlands, freshwater ecosystems, and all ecosystems through the Red List of Ecosystems.	Progress depends on existing and pending funding. Most projects are still in early stages, including requirements gathering, indicator exploration, dataset identification, and pilot analyses.	Prioritize projects according to funding status, readiness of datasets, indicators, and whether pilot studies have already been completed.
General constraints	Main constraints are funding and data availability. Datasets need to be collated, cleaned, and harmonized before use in assessments.	Final products must be easy for end users to access, use, and reuse. There is also a tension between user consultation needs and the time required to build infrastructure.	Develop an implementation pathway that combines funding strategy, data preparation, user consultation, and infrastructure design.

Topic	Key points	Constraints / challenges	Implications / actions
Reproducibility of methods and indicators	Core methodologies such as the Red List of Ecosystems are well documented, but reproducible indicators are often lacking.	Indicator execution can remain difficult because coding and programming skills vary among end users.	Provide clear workflows, reusable code, documentation, and user guidance to support reproducible indicator calculation.
Data findability	Data needed for indicators is often difficult to obtain, scattered, and poorly documented. Only a small proportion of participants found the search process moderately difficult.	Search processes vary widely, including search engines, online catalogues, and other sources.	Create clearer guidance on where to find relevant datasets and how to assess their suitability for indicators.
Data accessibility	Nearly half of participants (47%) reported that data was fully open to access, while 27% reported partial access requiring request or permission. The remaining participants reported restricted or inaccessible data.	Access barriers include complex access procedures, data-sharing agreements, difficult database navigation, data ownership issues, broken or outdated links, and low-quality metadata.	Prioritize open and well-documented datasets, while documenting access requirements and known barriers for restricted datasets.
How is the data accessed?	Most data are accessed directly from public repositories (40%) or extracted from published literature, such as supplementary data, printed tables, and figures (25%).	Smaller proportions of participants query specialized databases or use their own data. Downloading spatial data may be complicated.	Explore systems that reduce the need for users to download spatial data manually.

Topic	Key points	Constraints / challenges	Implications / actions
Data readiness & integration	Frequent challenges include varying data resolution and quality, inconsistent classification systems, spatial-scale differences, temporal mismatches, limited metadata, and missing or incomplete data.	These issues make it difficult to align datasets and compare indicators across ecosystems or regions.	Define minimum data standards, harmonization procedures, and metadata requirements for indicator workflows.
Skills required for data preparation	GIS and spatial analysis were reported by 35% of participants as required skills, while data cleaning and formatting were reported by 29%.	Ecological survey knowledge is important to ensure that data are used correctly, including understanding whether similar variables are measuring the same thing. Knowledge of species names, synonyms, and name changes is also important.	Support users with training, templates, and guidance covering spatial analysis, data cleaning, ecological assumptions, and taxonomy.
Reproducibility	Indicator calculation was considered partially reproducible by 67% of participants, while the remaining participants considered it difficult to reproduce.	Limits include a lack of documented processes and validation points, data availability, limited time to engage with managers, difficulty integrating expert knowledge, and variable data quality.	Strengthen documentation, validation checkpoints, data provenance, and mechanisms for incorporating expert knowledge.

Topic	Key points	Constraints / challenges	Implications / actions
Next steps	All participants highlighted stronger documentation as the priority of action for improving workflow reproducibility.	Important methodological details can be lost when workflows are only reported in scientific papers.	Develop complete documentation that captures assumptions, data sources, processing steps, code, validation checks, and decision points.

15 Example case study

15.1 Fleurieu Swamp (Fleurieu peninsula has 65 ecosystems - 23 wetlands and swaps +40 woodlands +2 marine zones)

Workflow stage	What happens	Reported challenge level
Prepare ecosystem distribution data	Global data sources needed reprojection/reconciliation with local expert layers; DCCEE spatial layers authoritative but differed in resolution from global sources.	Mixed
Prepare indicator variable data	BoM water balance data available but required preprocessing and temporal alignment with ecosystem snapshots.	Difficult
Calculation	Adapting thresholds from Red List (collapse) to Green Status Integrity/recovery was conceptually and technically challenging — different baselines and output formats.	Frustrating
Evaluating uncertainty	Expert review flagged as necessary but no formal protocol existed; validation data format mismatch meant manual reconciliation.	Uncertain

16 User Stories × Use Cases

! How to read these columns

A means the story and the use case share a requirement.

16.1 AD — Adaptability

User story	UC01	UC02	UC03	UC04	UC05	UC06	UC07	UC08
US-AD-01 As an ecologist, I want to adapt an existing indicator workflow to a new ecosystem type without rewriting the full analysis, so that I can reuse established methods and save weeks of rework. <i>Satisfies: AD03</i>								
US-AD-02 As a data analyst, I want to understand which parts of a workflow are ecosystem-specific vs generic, so that I know exactly what to change when switching context. <i>Satisfies: AD01</i>								
US-AD-03 As a government analyst, I want to switch the assessment purpose (e.g. risk to recovery) without needing a new workflow from scratch, so that I can serve multiple reporting needs with one underlying system. <i>Satisfies: AD04</i>								
US-AD-04 As an ecologist, I want to receive guidance on reconciling global and local data sources for the same ecosystem, so that my workflow is defensible, reproducible, and transparent. <i>Satisfies: AD05</i>								
US-AD-26 As an RSE, I want each workflow structured as numbered, self-contained notebooks (e.g. 01-data-download, 10-data-preparation, 20-calculate-indicator), so that individual steps can be re-run or replaced without re-running the entire pipeline. <i>Satisfies: AD01</i>								
US-AD-27 As an RSE, I want a single workflow template (Quarto .qmd or R Markdown .Rmd) enforcing a consistent structure (metadata block, data sources, methods, outputs, known limitations), so that all workflows are comparable and cataloguable. <i>Satisfies: AD03, RE01</i>								

User story	UC01C02C03C04C05C06C07C08
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US-AD-33 As an ecologist, I want workflows to calculate indicator values aligned to specific IUCN RLE criteria (A: extent decline; B: restricted distribution; C: abiotic condition; D: biotic condition), so that outputs are directly usable in risk assessment documentation.*Satisfies: AD04, VO01*

16.2 RE — Reproducibility and transparency

User story	UC01C02C03C04C05C06C07C08
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US-RE-05 As a data scientist, I want to run a workflow developed by someone else on a new dataset without modifying the code, so that I can validate and replicate published assessments independently.*Satisfies: RE06*

US-RE-06 As a research software engineer, I want to containerise the workflow environment with pinned dependencies, so that software versions don't break the workflow when R or packages change.*Satisfies: RE04*

US-RE-31 As a platform administrator, I want each workflow to declare its software environment (R/Python version, package list with versions) in a machine-readable format (renv.lock or requirements.txt), so that the environment is reproducible anywhere.*Satisfies: RE04*

US-RE-35 As a government reporting officer, I want each workflow run to produce a reproducibility record (timestamp, software environment snapshot, parameter values, input checksums), so that results can be independently verified.*Satisfies: RE01, RE02, RE03*

US-RE-38 As an RSE, I want each workflow registered in the indicator catalogue with a DOI and codemeta.json, so that workflows are citable as research software outputs.*Satisfies: RE07*

16.3 VO — Classification and vocabularies

User story	UC01UC02C03C04C05C06C07C08
US-VO-07 As an ecologist, I want to map local ecosystem classifications to global typologies (e.g. IUCN GET), so that my assessments align with international standards. <i>Satisfies: VO02</i>	
US-VO-08 As a data manager, I want a curated vocabulary mapping between national and global ecosystem classifications, so that others don't have to reconcile it themselves each time. <i>Satisfies: VO02, VO04</i>	
US-VO-09 As an indicator developer, I want vocabulary mapping to be machine-readable for workflow metadata, so that classification alignment can be automated rather than manual. <i>Satisfies: VO04</i>	
US-VO-10 As a policy officer, I want ecosystem types linked to authoritative identifiers (IBRA, IUCN GET IDs), so that reporting references are unambiguous and citable. <i>Satisfies: VO03</i>	
US-VO-37 As an RSE, I want workflows to use GET vocabulary identifiers for ecosystem types in all metadata fields, so that outputs are semantically interoperable with other Australian and global ecosystem datasets. <i>Satisfies: VO01, VO03</i>	

16.4 DD — Data discovery

User story	UC01UC02C03C04C05C06C07C08
US-DD-11 As a platform user, I want to know where ecosystem distribution data comes from and when it was last updated, so that I can assess its fitness for purpose before committing to an analysis. <i>Satisfies: DD02</i>	
US-DD-12 As an analyst working with restricted data, I want to run the workflow within the platform where the data lives, so that I can comply with data governance requirements while still doing the analysis. <i>Satisfies: DD05, AC04</i>	
US-DD-13 As a wetland ecologist, I want to integrate aerial survey data with satellite water extent time series, so that I can build a more complete picture of ecosystem condition. <i>Satisfies: DD04</i>	

User story	UC01UC02UC03UC04UC05UC06UC07UC08
US-DD-14 As a data scientist, I want guidance on transitioning from HPC preprocessing to cloud/local analysis, so that I'm not locked into one computing environment. <i>Satisfies: DD05</i>	
US-DD-15 As a marine ecologist, I want to apply consistent indicator calculations across regions with uneven survey coverage, so that results remain comparable despite data gaps. <i>Satisfies: DD03</i>	
US-DD-16 As an indicator developer, I want templates for harmonising heterogeneous survey datasets, so that I don't rebuild harmonisation logic for every new workflow. <i>Satisfies: DD03</i>	
US-DD-17 As a data analyst, I want clear documentation of data sources and known limitations embedded in each workflow, so that I can make informed fitness-for-purpose decisions before investing in an analysis. <i>Satisfies: DD06</i>	
US-DD-24 As an ecologist, I want to access named datasets (DEA data cube, NARcliM2.0, TASVEG, CRU TS) via documented API calls or download scripts, so that data provenance is captured from the first pipeline step. <i>Satisfies: DD01, RE02</i>	
US-DD-25 As a data scientist, I want an assessment check for each input dataset, so that I know which datasets are ready for use and which require transformation before entering a workflow. <i>Satisfies: DD02</i>	

16.5 DO — Documentation and training

User story	UC01UC02UC03UC04UC05UC06UC07UC08
US-DO-18 As a non-coder ecologist, I want step-by-step guidance with plain-language explanations at every stage, so that I can run the workflow without needing programming expertise. <i>Satisfies: DO01</i>	
US-DO-19 As a mid-career data scientist, I want documentation explaining what each workflow function does and why it exists, so that I can modify components confidently for new use cases. <i>Satisfies: DO04</i>	

User story	UC01C02C03C04C05C06C07C08
US-DO-20 As a senior scientist, I want documentation of how uncertainty is quantified and communicated in the outputs, so that I can defend the workflow's results to government decision-makers. <i>Satisfies: DO05, VA03</i>	
US-DO-21 As a workshop participant, I want worked examples covering common Australian ecosystem types, so that I can use the workflow as a template rather than starting from scratch. <i>Satisfies: DO02</i>	
US-DO-22 As a team leader, I want training materials that accommodate different learning levels within the same team, so that everyone can contribute regardless of technical background. <i>Satisfies: DO06</i>	
US-DO-23 As a platform administrator, I want workshops supported by published, reusable data and code outputs, so that participants have concrete materials to reference after the session ends. <i>Satisfies: DO03</i>	
US-DO-29 As an RSE, I want each workflow to include a data flow diagram (.dot or Mermaid) showing inputs, transformation steps, and outputs, so that I can understand the pipeline structure before running or modifying it. <i>Satisfies: DO04</i>	
US-DO-34 As a government reporting officer, I want indicator outputs delivered as standardised tabular summaries (CSV with a defined column schema) and spatial layers (cloud-optimised GeoTIFF), so that I can ingest them into reporting pipelines without reformatting. <i>Satisfies: DO07, AC05</i>	

16.6 AC — Accessibility

User story	UC01C02C03C04C05C06C07C08
US-AC-30 As an ecologist, I want to run a workflow on EcoCommons (cloud notebook) and locally (R/RStudio or Python) without modifying any code, so that I am not dependent on a single platform. <i>Satisfies: AC01, RE06</i>	
US-AC-32 As a platform administrator, I want compute and memory requirements specified per workflow, so that I can provision appropriate resources on Nectar or NCI without trial and error. <i>Satisfies: AC07</i>	

User story	UC01UC02UC03UC04UC05UC06UC07UC08
US-AC-39 As a government reporting officer, I want to access the catalogue and filter by ecosystem type (GET level), IUCN RLE criterion, or state/territory, so that I can identify available workflows for my reporting jurisdiction. <i>Satisfies:</i> AC06	

16.7 VA — Validation

User story	UC01UC02UC03UC04UC05UC06UC07UC08
US-VA-28 As an ecologist, I want workflows to validate input data against minimum quality thresholds (spatial coverage, completeness, temporal resolution) before running calculations, so that invalid inputs are caught early with informative error messages. <i>Satisfies:</i> VA01	
US-VA-36 As an ecologist, I want indicator values to include uncertainty estimates where methodologically appropriate, so that outputs communicate confidence for decision-making contexts. <i>Satisfies:</i> VA03, VA05	

16.8 Coverage

Category	Stories	Use case links	Use cases covered
Adaptability	7	32	8
Reproducibility and transparency	5	18	8
Classification and vocabularies	5	15	6
Data discovery	9	22	5
Documentation and training	8	30	7
Accessibility	3	11	6
Validation	2	8	5

Status

Pending review.

17 Requirements × Use Cases

17.1 AD — Adaptability

Requirement	UC0UC01UC02UC03UC04UC05UC06UC07UC08
AD01 Modular workflow designs that clearly separate ecosystem-specific components from generic analytical components.	
AD02 Clearly documented points within workflows where ecosystem definitions, data sources, indicator variables, and assessment purposes can be substituted.	
AD03 The ability to apply a common workflow structure across multiple ecosystem types.	
AD04 The ability to support alternative assessment frameworks, including ecosystem risk, ecosystem condition, recovery, and ecosystem accounting applications.	
AD05 Guidance for reconciling global datasets with local and jurisdictional data sources.	

17.2 RE — Reproducibility and transparency

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
RE01 Fully documented workflow processes from data acquisition through to final outputs.	
RE02 Clear provenance records for all input datasets.	
RE03 Version-controlled workflow code and analytical methods.	
RE04 Documented software dependencies and computational environments.	

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
RE05 Long-term maintainability as software platforms and packages evolve.	
RE06 Repeatable execution across multiple computing environments.	
RE07 Persistent identifiers and citation metadata for published workflows as research software outputs.	

17.3 VO — Classification and vocabularies

Requirement	UC01 UC02 UC03 UC04 UC05 UC06 UC07 UC08
VO01 Alignment with the IUCN Global Ecosystem Typology.	
VO02 Clear mapping between local ecosystem classifications and global typologies.	
VO03 Use of authoritative identifiers where possible.	
VO04 Machine-readable classification mappings and metadata.	
VO05 Transparent documentation of classification decisions and assumptions.	

17.4 DD — Data discovery

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
DD01 Guidance on identifying authoritative ecosystem and indicator datasets.	
DD02 Clear documentation of data provenance, quality, currency, and fitness-for-purpose.	
DD03 Consistent approaches for harmonising heterogeneous datasets.	
DD04 Mechanisms to support integration of biological surveys, monitoring data, remote sensing products, and spatial datasets.	
DD05 Support for workflows operating across local, cloud, and high-performance computing environments.	

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
DD06 Explicit documentation of data limitations and known biases.	
DD07 Approaches that reduce the effort and time required for data preparation relative to manual processing.	

17.5 DO — Documentation and training

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
DO01 Plain-language workflow documentation.	
DO02 Worked examples using real ecosystem assessments.	
DO03 Step-by-step implementation guidance.	
DO04 Technical documentation describing workflow functions and analytical methods.	
DO05 Guidance on interpretation of outputs, assumptions, limitations, and uncertainty.	
DO06 Training resources suitable for beginner, intermediate, and advanced users.	
DO07 Standardised output formats and schemas so indicator products can be ingested without reformatting.	

17.6 AC — Accessibility

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
AC01 Support for commonly used open-source tools and platforms.	
AC02 Use of technologies already familiar to the target user community where possible.	
AC03 Clear installation and setup instructions.	
AC04 Support for users operating within institutional security and software restrictions.	
AC05 Outputs that can be interpreted by both technical and non-technical users.	

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
AC06 A searchable workflow catalogue supporting filtering by ecosystem type, assessment criterion, and jurisdiction.	
AC07 Documented compute and memory requirements for each workflow to support resource provisioning.	

17.7 VA — Validation

Requirement	UC01UC02UC03UC04UC05UC06UC07UC08
VA01 Explicit validation procedures.	
VA02 Documentation of assumptions and methodological limitations.	
VA03 Quantification and communication of uncertainty.	
VA04 Support for expert judgement where empirical validation data are unavailable.	
VA05 Consistent reporting of confidence in indicator outputs.	
VA06 Documented review protocols specifying how and when expert input is sought, including fallbacks where experts are unavailable.	

17.8 Coverage by category

Category	Requirements	Use case links	Use cases covered
Adaptability	5	17	6
Reproducibility and transparency	7	24	8
Classification and vocabularies	5	13	6
Data discovery	7	14	5
Documentation and training	7	22	7
Accessibility	7	15	8
Validation	6	18	5

 Status

Pending.